Dvoretzky covering problem for general measures

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Random covering sets

Let μ be a Borel probability measure on $\mathbb{R}^d$ and

- Fix a sequence of positive real numbers $\underline{r} = (r_n)_n$, with $r_n \downarrow 0$.
- Pick a sequence of points $\omega = (\omega_n)_n$, i.i.d with distribution μ.

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Definition

The random covering set is the set

$$E_{\underline{r}} = E_{\underline{r}}(\omega) = \{x \in \mathbb{R}^d : x \in B(\omega_n, r_n) \text{ i.o.}\} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} B(\omega_n, r_n).$$

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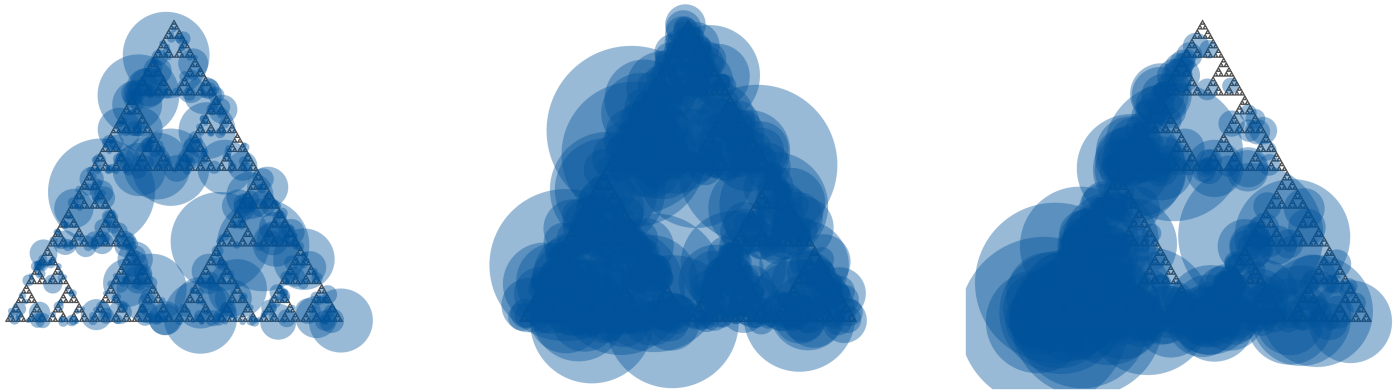
Basic question

How large is $E_{\underline{r}}$ for a typical (w.r.t. $\mathbb{P}_\mu = \mu^{\mathbb{N}}$) realisation?

Dvoretzky covering problem



(a) $\underline{r} = (n^{-\frac{4}{5}})_{n=3}^{503}$
μ =natural measure

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Figure: 500 random balls on the Sierpinski triangle

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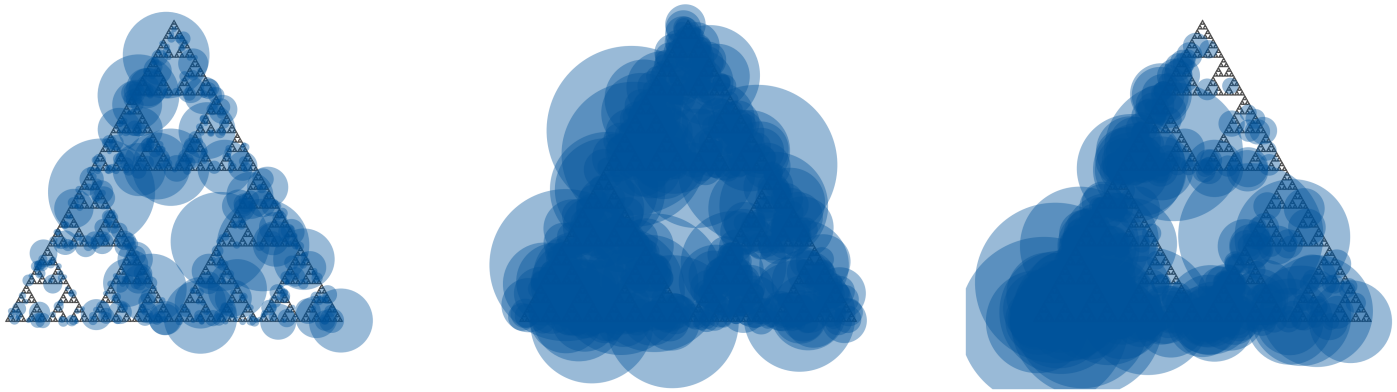

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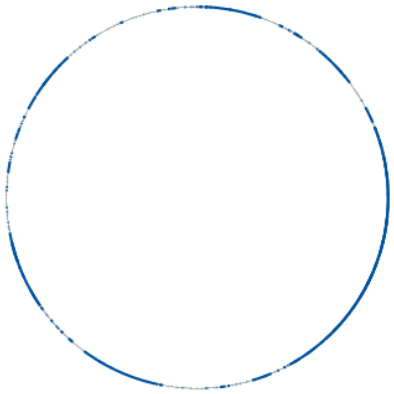
Dvoretzky covering problem

Let μ be a Borel probability measure on $\mathbb{R}^d$ and let $A \subset \mathbb{R}^d$ be measurable. When is $A \subset E_{\underline{r}}$, $\mathbb{P}_\mu$-almost surely.

Note: Always $\mathbb{P}_\mu(A \subset E_{\underline{r}}) \in \{0, 1\}$ by Kolmogorov's zero-one law.

Random covering sets: History

Classical case: $\mu = \mathcal{L}$, the Lebesgue measure on the one dimensional torus $\mathbb{T}$.



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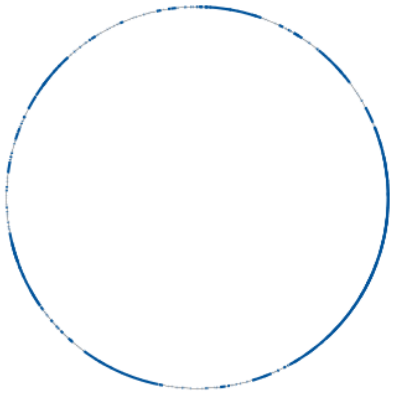

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Observation (Borel 1897)

For a fixed $x \in \mathbb{T}$, we have

1. $x \notin E_{\underline{r}}$ almost surely, if $\sum_{n=1}^{\infty} r_n < \infty$,
2. $x \in E_{\underline{r}}$ almost surely, if $\sum_{n=1}^{\infty} r_n = \infty$.

Follows from the Borel-Cantelli lemma, since

$$\mathbb{P}(x \in B(\omega_n, r_n)) = \mathbb{P}(\omega_n \in B(x, r_n)) = \mathcal{L}(B(x, r_n)) = 2r_n,$$

and since $\sum_{n=1}^{\infty} 2r_n = \infty$ if and only if $\sum_{n=1}^{\infty} r_n = \infty$.

Random covering sets: History

By Fubini's theorem, we have

$$\begin{aligned}\mathbb{E}(\mathcal{L}(E_{\underline{r}})) &= \iint \chi_{E_{\underline{r}}}(x)\,\mathrm{d}x\,\mathrm{d}\mathbb{P} \\ &= \iint \chi_{E_{\underline{r}}}(x)\,\mathrm{d}\mathbb{P}\,\mathrm{d}x \\ &= \int \mathbb{P}(x \in E_{\underline{r}})\,\mathrm{d}x,\end{aligned}$$

so Borel's observation shows that, almost surely,

$$\mathcal{L}(E_{\underline{r}}) = \begin{cases} 0, & \text{if } \sum_n r_n < \infty \\ 1, & \text{if } \sum_n r_n = \infty. \end{cases}$$

- This gives a complete description of the size of $E_{\underline{r}}$ in terms of its Lebesgue measure.
- Opens the way for interesting follow-up questions: If $\sum_n r_n < \infty$ what is the (Hausdorff) dimension of $E_{\underline{r}}$?

Dvoretzky covering problem: History

Dvoretzky 1956: Does the condition $\sum_{n=1}^{\infty} r_n = \infty$ imply that $\mathbb{T} \subset E_{\underline{r}}$ almost surely?

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Theorem (Dvoretzky 1956)

1. *If $r_n \geqslant \frac{\log n}{n}$ for all large enough n, then $\mathbb{T} \subset E_{\underline{r}}$, almost surely.*

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2. *There exists a sequence $\underline{r} = (r_n)_n$, such that $\sum_{n=1}^{\infty} r_n = \infty$, but $\mathbb{T} \not\subset E_{\underline{r}}$, almost surely.*

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Borel's observation and Dvoretzky's result have the following corollary for polynomially decreasing sequences of radii $\underline{r} = (cn^{-t})_n$:

Corollary

Let $\underline{r} = (cn^{-t})_n$, with $c > 0$.

1. *If $t > 1$, then $\mathbb{T} \not\subset E_{\underline{r}}$ almost surely.*
2. *If $t < 1$, then $\mathbb{T} \subset E_{\underline{r}}$ almost surely.*

Dvoretzky covering problem: History

For the critical exponent $t = 1$, the covering property depends on the constant c:

Theorem

1. *(Kahane 1959) If $r_n = cn^{-1}$, for $c > \frac{1}{2}$, then $\mathbb{T} \not\subset E_{\underline{r}}$ almost surely.*

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3. *(Erdös 1960, Mandelbrot 1972) If $r_n = \frac{1}{2}n^{-1}$, then $\mathbb{T} \subset E_{\underline{r}}$ almost surely.*

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The Dvoretzky covering problem for covering the full torus was solved by Shepp in 1972:

Theorem (Shepp 1972)

If $\underline{r} = (r_n)_n$, then $\mathbb{T}$ is covered almost surely by $E_{\underline{r}}$ if and only if

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \exp\left(\sum_{k=1}^{n} 2r_k\right) = \infty.$$

Capacity and coverings

Let $\Phi_{\underline{r}} \colon \mathbb{T}^2 \to \mathbb{R}$ denote the kernel function

$$\Phi_{\underline{r}}(x,y) = \exp\left(\sum_{n=1}^{\infty} \max\{2r_n - |x-y|, 0\}\right) = \exp\left(\sum_{n=1}^{\infty} \mathcal{L}(B(x,r_n) \cap B(y,r_n))\right).$$

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$$\mathrm{Cap}_{\underline{r}}(A) = \sup\{I_{\underline{r}}(\nu)^{-1} \colon \nu \text{ is a Borel probability measure on } A\}.$$

Here we interpret $\infty^{-1} = 0$, so $\mathrm{Cap}_{\underline{r}}(A) = 0$ if and only if $I_{\underline{r}}(\nu) = \infty$ for all measures ν on A.

Capacity and covering

Kahane proved the following:

Theorem (Kahane 1990)

A compact set $C \subset \mathbb{T}$ is covered almost surely by $E_{\underline{r}}$ if and only if

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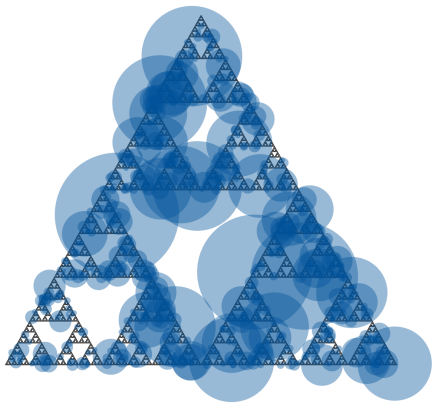
- Note: One can show that $\mathrm{Cap}_{\underline{r}}(\mathbb{T}) = 0$ if and only if $I_{\underline{r}}(\mathcal{L}) = \infty$, i.e. the energy of the Lebesgue measure characterises the covering property.
- This is essentially how Shepp's proof works.

Random covering sets: General setting



For random covering sets driven by arbitrary measures μ on $\mathbb{R}^d$, basic results work essentially in the same way as for the Lebesgue measure:

Lemma

For any $x \in \mathbb{R}^d$, we have

1. $x \notin E_{\underline{r}}$ $\mathbb{P}_\mu$*-almost surely, if* $\sum_{n=1}^{\infty} \mu(B(x, r_n)) < \infty$,
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- Again follows from Borel-Cantelli, since $\mathbb{P}_\mu(x \in B(\omega_n, r_n)) = \mathbb{P}_\mu(\omega_n \in B(x, r_n)) = \mu(B(x, r_n))$

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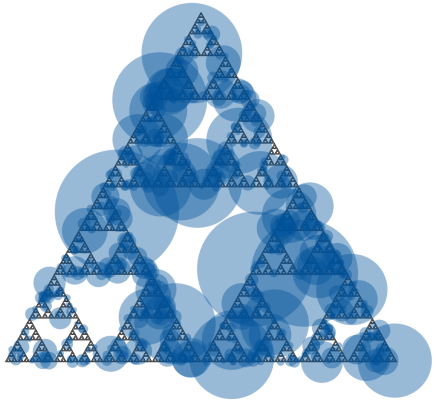

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- Using Fubini, if $A \subset \mathbb{R}^d$, $\sum_{n=1}^{\infty} \mu(B(x, r_n)) = \infty$ for all $x \in A$ and ν is a Borel probability measure supported on A, then $\nu(E_{\underline{r}}) = 1$, $\mathbb{P}_\mu$-almost surely.

Dvoretzky covering problem: General setting

For measures other than $\mathcal{L}$, a full characterisation for the Dvoretzky covering problem is only known in the following cases:

- (Fan-Karagulyan 2021) For $\mu = f \, \mathrm{d}\mathcal{L}$, under some reqularity assumptions for f.
- (Kahane 1990) For $\mu =$ Lebesgue measure on $\mathbb{T}^d$, but for simplices (homothetic triangles) instead of balls.

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For $\underline{r} = (cn^{-t})$ the critical exponent is known for

- (Tang 2012) Fully supported Borel probability measures on $\mathbb{T}$,
- (Seuret 2018) Gibbs measures on irreducible topological Markov shifts (homogeneous self-similar measures).

At the critical exponent, the critical constant is not known in any non-Lebesgue case.

Capacity and covering

For $\mathcal{L}$ define the kernel function $\Phi_{\underline{r}}\colon \mathbb{T}^2 \to \mathbb{R}$ by

$$\Phi_{\underline{r}}(x,y) = \exp\left(\sum_{n=1}^{\infty} \mathcal{L}(B(x,r_n)\cap B(y,r_n))\right),$$

and then the $\underline{r}$-energy of a measure ν by

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Main result

For a Borel probability measure μ we let

$$X_\mu = \left\{ x \in \operatorname{spt} \mu \colon \sum_{n=1}^{\infty} \mu(B(x, r_n))^2 < \infty \right\}$$

Main result

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Theorem (A.-Myllyoja 2026+)

Let μ be a Borel probability measure on $\mathbb{R}$, and let $A \subset \mathbb{R}$ be analytic. Then $A \subset E_{\underline{r}}$, $\mathbb{P}_\mu$-almost surely, if and only if

$$\mathrm{Cap}_{\mu,\underline{r}}(A \cap X_\mu) = 0.$$

Remarks

- Same result holds for measures on $\mathbb{T}$ instead of $\mathbb{R}$.
- By working in $\mathbb{R}$ instead of $\mathbb{T}$ we lose two properties which made earlier methods simpler: Compactness of the support of μ and rotational invariance.
- Even for $\mu = \mathcal{L}$ the result is new for analytic sets.

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- The "only if" part of the theorem—$\mathrm{Cap}_{\mu,\underline{r}}(A \cap X_\mu) > 0$ implies that A is not covered almost surely—is well known for arbitrary measures on $\mathbb{R}^d$.
- The major difficulty is in proving that $\mathrm{Cap}_{\mu,\underline{r}}(A \cap X_\mu) = 0$ implies covering.
- While $I_{\underline{r}}(\mathcal{L}) = \infty$ characterises the covering of $\mathbb{T}$, this is not true in general: There are examples of "natural measures" on sets where the energy of the natural measure does not characterise the covering of the set.
- One such example is when μ is the Hausdorff measure on the Cantor set C: There are sequences $\underline{r}$, where $I_{\mu,\underline{r}}(\mu) = \infty$, but $\mathrm{Cap}_{\mu,\underline{r}}(C) > 0$.

On the proof

First difficulty: How to use the condition $I_{\mu,\underline{r}}(\nu) = \infty$ for all Borel probability measures ν supported on $A \cap X_\mu$ to get get information on the covering property of the set $A \cap X_\mu$.

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Proposition (A.-Myllyoja 2026+)

Let μ be a Borel probability measure on $\mathbb{R}^d$ and $A \subset \mathbb{R}^d$ be analytic. If for all Borel probability measures ν supported on A, there exists a Borel set $A' \subset A$, such that $\nu(A') = 1$, and $A' \subset E_{\underline{r}}$ $\mathbb{P}_\mu$-almost surely, then $A \subset E_{\underline{r}}$ $\mathbb{P}_\mu$-almost surely.

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Important

Recall that $\sum_n \mu(B(x, r_n)) = \infty$ for all $x \in A$ implies that $\nu(E_{\underline{r}}(\omega)) = 1$ for $\mathbb{P}_\mu$-almost every ω. However, the full measure set that is covered can depend on the realisation of the process! The observation does not say that exists a set $A' \subset A$, with $\nu(A') = 1$, such that $A' \subset E_{\underline{r}}(\omega)$, for $\mathbb{P}_\mu$-almost every ω.

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Important

Recall that $\sum_n \mu(B(x, r_n)) = \infty$ for all $x \in A$ implies that $\nu(E_{\underline{r}}(\omega)) = 1$ for $\mathbb{P}_\mu$-almost every ω. However, the full measure set that is covered can depend on the realisation of the process! The observation does not say that exists a set $A' \subset A$, with $\nu(A') = 1$, such that $A' \subset E_{\underline{r}}(\omega)$, for $\mathbb{P}_\mu$-almost every ω.

Second difficulty: Use the condition $I_{\mu,\underline{r}}(\nu) = \infty$ for a fixed measure ν to construct a deterministic set A' of full measeure which is covered almost surely. This is where the assumption that μ is supported on $\mathbb{R}$ is used!

On the proof

To pass from covering $A \cap X_\mu$ to covering A, we have the following lemma.

Lemma

Let μ be a Borel probability measure on $\mathbb{R}$. Let $\varepsilon > 0$ and $A \subset \mathbb{R}$ be an analytic set which satisfies

$$\sum_{n=1}^{\infty} \mu(B(x, r_n))^{1+\varepsilon} = \infty,$$

for all $x \in A$. Then $A \subset E_{\underline{r}}$ almost surely.

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- Proof is elementary and does not use any machinery needed in the proof of the main theorem.
- Recall that for $\varepsilon = 0$, the condition only implies $\nu(E_{\underline{r}}) = 1$ almost surely for all measures ν on A.

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- Proof is elementary and does not use any machinery needed in the proof of the main theorem.
- Recall that for $\varepsilon = 0$, the condition only implies $\nu(E_{\underline{r}}) = 1$ almost surely for all measures ν on A.
- The lemma shows that $A \setminus X_\mu = \{x \in A \colon \sum_{n=1}^{\infty} \mu(B(x, r_n))^2 = \infty\}$ is covered almost surely automatically.

On the proof

The lemma has a straightforward corollary:

Corollary (A.-Myllyoja 2026+)

Let μ be a Borel probability measure on $\mathbb{R}^d$, $A \subset \mathbb{R}^d$ be analytic, $c > 0$ and $\underline{r} = (cn^{-t})_n$.

1. *If $t > (\sup_{x \in A} \underline{\dim}_{\mathrm{loc}}(\mu, x))^{-1}$, then $A \not\subset E_{\underline{r}}$, $\mathbb{P}_\mu$-almost surely.*
2. *If $t < (\sup_{x \in A} \underline{\dim}_{\mathrm{loc}}(\mu, x))^{-1}$, then $A \subset E_{\underline{r}}$, $\mathbb{P}_\mu$-almost surely.*

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- This result was previously known for fully supported measures on $\mathbb{T}$ (Tang 2012) and Gibbs measures for irreducible Markov shifts (Seuret 2018).
- For the normalised s-Hausdorff measure on strongly separated self-conformal sets with Hausdorff dimension s, we can characterise the critical constant c for the covering problem with the critical exponent $t = \frac{1}{s}$.
- The latter characterisation needs the full power of our main result.

Thank you! Questions?